Release Notes

CADP internal QA build release notes — generated from OpenProject. Not the in-app Release Notes feature used by tenants.

1. Release Overview

Field	Details
Product Name	**CADP Workspace**
Release Version	**Release 1.0.0 (29)**
Release Date	**November 11, 2026**
Release Type	**Feature & Bug Fix Release**
Prepared By	**Development Team**
Approved By	**Ganagavathy**
Environment	**QA**

2. Release Summary

This release of CADP Workspace Version 1.0.0 (29) delivers 34 bug fixes, 3 new features, and 10 enhancements across the platform. Highlights include A formula returning a list no longer fails silently, and DISTINCT_* works, A formula field now means the same thing wherever the write came from, and Validate Expression warns when a formula returns a list. Refer to the Known Issues section for current limitations.

3. New Features

Feature ID	Feature Name	Description	Impact
	A formula returning a list no longer fails silently, and DISTINCT_* works	Three related changes to how the engine handles a formula that produces many values; rather than one. Raised by QA, who read an empty field as a broken formula.; 1. **A list-returning formula is now called out.** `UNIQUE()`, and any aggregate given a; groupBy, returns a list; a formula field shows one value, so the field renders blank.; Validate Expression now shows a warning saying so, and evaluate-form-batch returns `null`; plus an `errors[]` entry naming how many values came back. Previously the field simply went; blank with `errors: []` - indistinguishable from a formula that legitimately produced; nothing. The formula still SAVES: it is not wrong, it just cannot be displayed.; 2. **An aggregate handed a list is refused.** `COUNT_RECORDS(UNIQUE({price}))` reads like; "count the distinct prices" and answered 0. The field token is quoted only for the OUTER; aggregate, so the nested call received `{__aggN}` unquoted, resolved nothing and returned an; empty list for the outer one to count. It is refused at validate time, naming the; `DISTINCT_*_OF_RECORDS` function to use instead. Deliberately NOT made to work: counting a; list would give COUNT_RECORDS two meanings over the same data (4 records, 2 distinct) and; leave its filter argument ambiguous.; 3. **The DISTINCT_* family now works at all.** Two faults, the first hiding the second.; `isInsideAggregateCall` and `isAggFilterArg` matched six names, and; `DISTINCT_COUNT_OF_RECORDS` contains `COUNT_OF_RECORDS`, not `COUNT_RECORDS`, with no word; boundary after `DISTINCT_` - so none could ever match it, and its field token was treated as; an ordinary single-record lookup that matched every record: "Multiple records found". Past; that, the evaluator read only `__featureRecords__`, while the resolver had already fetched; the values and substituted them as `{__aggN}` - the way the PLAIN aggregates receive them -; so it answered 0. It now reads that array, de-duplicates, and applies the same metric; (hoisted to a shared `distinctApply` rather than duplicated).	

Feature ID	Feature Name	Description	Impact
	A formula field now means the same thing wherever the write came from	one shared recalculation service (`formulaProjection.service.js`) used by every; path that writes a record, plus an internal endpoint the workflow service calls after it; writes.; Before this, which fields a write recalculated depended on the door it came through:; | Write path | Where formula fields got their value | Correct for the whole language? |; |---|---|---|; | Manual create / submit | `unified-event-processor.js` → `applyProjectionFormulas` | yes |; | Bulk import | `importFormula.service.js` → `applyProjectionFormulas` | yes |; | Direct field update | `featureData.service.js` → inline `new Function` | **no — arithmetic only** |; | Workflow record write | nothing | **no — never ran** |; The `new Function` path substituted `{f:uuid}` with numbers and only evaluated when the; result was purely numeric and arithmetic, so `IF(...)`, `ROUND(...)`, `LOOKUP(...)` and every; aggregate silently produced nothing on a direct update (view-cart quantity increment, inline; grid edits) while the same expression worked when submitted. The workflow path wrote straight; to the collection with the driver, so a formula field it touched kept the node's value only; until the record next opened in create mode, where the form recomputed over it.	
	Validate Expression warns when a formula returns a list	`UNIQUE()`, and any aggregate given a groupBy, returns a list of values. A formula; field shows one value, so such a formula renders blank. Validate Expression now shows an amber; warning saying exactly that, instead of the green "Formula is valid" that gave no hint why the; field would stay empty.	

4. Enhancements / Improvements

Enhancement ID	Module	Description	Benefit
	data-import	The batch import (`/data/batch-import/upload`) now accepts `lookupPolicy`. In each; 10,000-row batch, rows with missing lookup values wait until after the first lookup pass. All; missing values in the batch are then created with one `importFeatureDataWithMode` call per; referenced feature, with required-field Default Values applied and no-default rows skipped. Only; the waiting rows are checked again, with a fresh lookup cache. A value that fails to create; reports why on its row. `referenceCreated` is stored on the batch job and returned by its status; API. The policy parser and the job-counter helper moved to `src/utils/importLookupPolicy.js`, so; both pipelines use the same code.	Lookup features now import through the batch pipeline, which ignored the; Referenced Feature Import box.
	data-import	When an import auto-creates a missing lookup value in the referenced feature, any; required field of that feature that is not in the file now takes its configured Default Value.; If a required field has no Default Value, the record is not created and the row is skipped with; "Record not created in referenced feature "X" for "V": required field(s) A, B have no Default; Value set, so this row was skipped." Unique, auto-generated and session-param fields are never; filled from a default. The import status now returns `referenceCreated`; (`[{ featureId, featureName, count, defaultsApplied[], skipped, missingDefaults[] }]`), totalled; across chunks. Applies to the chunked import (Pipeline A), for lookup fields and reference-table rows.	Auto-create failed on every referenced feature that had any other required field.

Enhancement ID	Module	Description	Benefit
	data-import	On the batch import path, a matched row whose file values already equal the stored record; (e.g. Country Code IND, Currency INR → stored INR) is no longer written or counted as Updated. It is; counted in a new `unchangedCount` (job doc, `/batch-import/:jobId/status`) and the Import modal shows; "Same record, no changes: N" under Created / Updated; "Successfully Imported" includes it.; Comparison is `isPatchUnchanged` (utils/batchImportMatch.js): values compare as text (`10` = `"10"`),; objects by JSON, encrypted fields by the stored `_hash` (missing hash = changed). A row carrying; reference-table rows always counts as changed. Unchanged rows are not sent to onUpload workflows.	Re-importing an exported file reported every matched row as Updated even when nothing changed.
	data-import	Features whose dropdown is sourced from another feature's records (e.g. Inventory →; Product Name) now import through the batch pipeline (10,000 rows per write) instead of the; chunked one; only features with a table still use the chunked pipeline. Lookup resolution now; queries each distinct value once per chunk / batch instead of once per row (`findLookupMatches`; memoizes hits in the existing per-chunk/per-batch context cache, keyed by target feature + built; query clauses). Misses are not cached, so a record auto-created mid-import is still found on; re-inspect.	Importing such a feature sat at 0% for a long time — the chunked path reports progress; only per finished 100-row chunk, and each row ran its own unindexed lookup query plus a per-row save.
	data-import	Importing a feature with a lookup/reference dropdown is now about as fast as one without.; Before the per-row loop, `prefetchLookupMatches` collects each lookup field's distinct values,; fetches the matching source records with one `$in` query per 2,000 values; (`LOOKUP_PREFETCH_VALUES_PER_QUERY`), pairs them in memory (strict equality, array fields match; by containment, same as MongoDB) and seeds the per-batch cache `findLookupMatches` reads. Row; results, ambiguity (2+ matches) and error messages are unchanged. Unsupported clause shapes; (dotted/operator aliases) and query errors fall back to the per-row path. Misses are cached unless; lookup auto-create is on. Used by both the batch and the chunked pipeline.	20,000 rows with 13,166 distinct Product Names meant ~13,000 unindexed scans of the; source feature, one per row. Measured after the change: 20,000 rows imported in 24 s.

Enhancement ID	Module	Description	Benefit
	feature-data-export	Layout/display components (e.g. "2 Column Grid", type `column-grid`) no longer appear as empty; columns in exported files. Field defs now skip every type in the shared; `NON_ATOMIC_COMPONENT_TYPES` set (tables exempted — export handles them separately).; Excel exports: the header row of every sheet (Records + table detail sheets) is locked via sheet; protection; data cells and new rows stay editable (insert/delete rows, sort, filter allowed), so the; file can still be edited and re-imported. No password — guards against accidental header edits only.	The export's local blocklist only had `grid`/`column`/`row`…, not `column-grid`, so grids; leaked into the header row. Header lock requested to keep the import-matching headers intact.
	component-properties	`ADVANCED_SECTION_VISIBILITY` lists `'Interactions': ['*']` (last, so it is added after every; other Advanced section). New `ADVANCED_SECTION_FIELDS` gives a section added through that list its fields; (only Interactions has any; every other section is still added with `fields: []`). The fields live in the; new `src/seeders/interactionSectionFields.js`: ≈235 entries `{ key, label, type: 'custom' }`, some with; `options: [{ value, label }]` naming the saved values of a dropdown. `{name}` in a label is filled by the; builder. No `defaultValue` on any of them (they are words, not settings — the table column drop copies; field defaults into `componentProps`).; `withEveryTypeSections` also adds such a `'*'` section (last) to each type's own document in the base-doc; pass: a type missing from the components list (e.g. `image`, `select`, `row`, the charts) never reaches; `buildMergedGroups`, and the API adds only `__common__` to it — the first seed run left ≈100 types; without Interactions, which the builder used to force onto every component. Dry-run of the whole seeder; (Mongo stubbed) against the previous version: all 144 written types identical apart from the added; Interactions; every type has it, last. Not touched: 54 leftover documents of types no longer seeded or in; the palette (July–September).	Interactions was the only Advanced section the builder added by itself, with its words hard-coded.; Now it is listed, and worded, by the component properties like the rest.

Enhancement ID	Module	Description	Benefit
	data-import	Restored the "Referenced Feature Import" checkboxes in `ImportTransactionsModal`. The; chunked upload sends `lookupPolicy` again, so ticking a box really creates missing values. The; import summary now has a "Referenced features" section under Created / Updated / Same record:; "Created in X: n", which required fields got their Default Value, and "Skipped in X: n" with the; required fields that have no Default Value.	Auto-creating records in referenced features is needed again.
	formula-field-properties	`DISTINCT_COUNT_OF_RECORDS()` and its four siblings now appear in the Aggregate; category of the formula builder's **Operators and Functions** tab, beside the five that were; already there.	that tab holds its own hardcoded list, separate from autocomplete and from Help. The; DISTINCT family has never been in it - checked with `git log -S`, this is not a removal - so the; only ways to reach those functions were typing the name and accepting the completion, or reading; Help. Anyone browsing the list to find "count the distinct values" found nothing and concluded; the platform could not do it.

Enhancement ID	Module	Description	Benefit
	interaction-properties	**The section is listed by the component properties, like every other Advanced section.** The builder used; to add "Interactions" itself to every component (`[...apiAdvancedOrder, 'Interactions']`, an; `Interactions: {}` key, `'Interactions': ['*']` in the frontend visibility list, and two; `sectionName === 'Interactions'` exceptions). All removed: the rbac seeder now lists it for every; component type, last, so it shows in the same place as before.; **Every word of the panel is read from that section** (≈235 fields: labels, hints, placeholders, empty /; loading / error messages, the names in its dropdowns, and the notes above the panel for a carousel; slide, navbar item, listing button, navbar container and listing card). Covers InteractionProperties,; PassDataFieldMap, ShopperChoiceActionSection, ShowOnlyPartsPicker, ListingBarActionInteractions and the; notes in advancedFieldsProp. `interactionWords.tsx` gives them the words (`InteractionWordsProvider`,; `useInteractionWords`, `readInteractionWords`): `t(key, values)` fills `{name}`, `rich` places nodes; (bold names), `optionLabel(key, value)` names a saved value. A word the section lacks shows as nothing —; never a made-up text.; **The code keeps the values and the steps; the section keeps the words.** Dropdown values (`onClick`,; `navigate`, `eq`, `sm`, …) are unchanged and saved as before; only their shown names come from the; section's `options`. Problem messages under a mapping are now picked by key (`issueNoMaster`, …).; Dropdowns that used to show raw code words now show names: Trigger (Click, Double click, …), Action; (Navigate, Open popup, Save shopper choice, …), Navigation Target, Feature Page Mode, Record Source.	Interactions was the only Advanced section not defined in the database — added by code to every; component, with ~320 hard-coded words. Now it follows the same pattern as the rest, and its words are; changed in the seed, not in code.

5. Bug Fixes

Bug ID	Description
1217	Dark mode is not applied consistently across all application screens and UI components.

Bug ID	Description
1236	Components → Table → Data Table Properties → Data Source → Platform Feature →S.No. column is blank when loading data from a Platform Feature.
1240	Cloned feature shows inconsistent publish status, preventing record creation.
1412	Publish to Environment : Cloned module is not listed under Changes in the Publish Workspace dialog.
1423	Workspace → Portal Management→ Edit, Duplicate, and Delete options are not working in the Portal Management menu.
1428	Workspace → Portal Management → Roles & Permissions→ Edit and Delete options are not working in the Roles & Permissions menu.
1430	Workspace → Portal Management → Portal Access→ Duplicate spinner controls are displayed for the Session Timeout field.
1431	Workspace → Portal Management → Portal Access → Invite Only→ Duplicate spinner controls are displayed for the Invite Expiry (Hours) field.
1432	Workspace → Portal Management → Portal Users→ Three-dot menu options are hidden in Portal Users.
1433	Workspace → Portal Management → Portal Users → Create User→ Phone Number field accepts invalid input in Create User.
1435	Workspace → Portal Management → Portal Users→ Edit and Remove options are not working in Portal Users.
1440	Workspace → Portal Management → Portal Users→ Copy, Export, and Import buttons are not working in Portal Users.
1507	Vertical Template → Creating a Vertical Template fails during data submission with a 500 Internal Server Error displaying "Feature is not defined."
1508	Feature Data Import : Generated import template incorrectly validates ItemCode as a dropdown with values Active and Inactive.
1554	Workspace Management → Roles & Permission →No 'No Data Found' Message Displayed for Invalid Role Search
1558	Selected Default Workspace Is Not Automatically Loaded After User Logs Back Into the Application
1623	Formula-->The SUM_OF_RECORDS() formula field does not calculate or display the total sum of records when used with & without filter conditions. Instead, it displays 0.00 even though valid numeric records are available.

Bug ID	Description
1624	Formula-->The COUNT_RECORDS() formula returns 0.00 instead of the correct record count when valid records are available and no filter conditions are applied.
1628	Formula-->The AVG_OF_RECORDS() formula returns 0.00 instead of the correct average value, both with and without applying filter conditions, even when valid numeric records are available.
1632	Formula-->The MIN_OF_RECORDS() formula returns 0.00 instead of the correct minimum value, both with and without applying filter conditions, even when valid numeric records are available.
1636	Formula-->The MAX_OF_RECORDS() formula returns 0.00 instead of the correct maximum value, both with and without applying filter conditions, even when valid numeric records are available.
1638	Formula-->The UNIQUE() formula field returns a blank output even when the mapped Department field contains valid values.
1770	User Management->New User->Platform options are not displayed when creating a user in a newly created workspace
1772	Workspace Management → User Management →User Management displays users from the previous workspace when accessing a newly created workspace
1775	Role Management→ Super Admin permissions are not automatically assigned in Role Management for newly created workspaces
1795	Formula-->Temp variable formula does not calculate the expected value in the Final Order Amount field.
1812	Portal Management → When I try to configure the module, the created module does not appear below the dropdown.
1849	Vertical Template-->Vertical Template creation displays only one module instead of all modules from the selected existing workspace.
1931	workspace-->Application name validation incorrectly counts a space as a character. When 2 characters and 1 space are entered, clicking Create does nothing and no validation message is displayed.
1932	Workspace-->The module and feature are already published, but the feature table list page incorrectly shows the status as Unpublished.
2064	Reset-password OTP has no attempt limit
2079	Role & Permission → Export → Created By and Created At Fields Missing in Excel

Bug ID	Description
2080	User Management →The application incorrectly uses the first three digits of the user's mobile number as the PIN Code when creating a user through the Feature Properties configuration.
2081	In Jason Page Close Button Overlaps with AI Tab.

6. Configuration Changes

Configuration	Description	Impact
None	No configuration changes are required for this release.	N/A

7. API Changes

API Name	Type	Description
None	N/A	No API changes are included in this release.

8. Known Issues

Issue ID	Description	Workaround
	Role Management Change Effect	Because some menu structures have changed, certain features may return a 403 status code. To resolve this, edit the role, select the platform feature, and save the role again.
	Scheduler with Workflow Not Working	Use the conventional feature-wise approach wherever workflow scheduling is required.
	Formula Builder UI – Field Insertion Through Typing	This works in current fields. For external feature fields, it won't work, as there could be fields with same names from multiple features.
	Formula Builder – AI-Based Formula Writing/Editing Not Implemented	Use manual formula entry instead.
364	Organization Details → Basic Info: Website URL accepts invalid URL formats without validation.	Expected: Validate the URL format and show an error message for invalid values, such as www.example or example@123.www.example

Issue ID	Description	Workaround
	Builder Tool Not Available on Mobile Web	The builder tool is not supported on mobile web. This is implemented as expected — builder functionality is restricted to desktop browsers by design. It should be available on Tab though.
	Mobile – Formula Field Not Working in Tables & Charts	Formula fields are not evaluated correctly when used inside Tables and Charts on mobile. This is under review.

9. What to Test

- Login – Login, Sign In with Google (have to be configured in external providers in Tenants of metadata), Sign In with Microsoft (have to be configured in external providers in Tenants of metadata)
- **Create Application**
    1. Start from Scratch
    2. Choose/Create Vertical Template

- **AI-Assisted Creation & Ask FAQ** (Cavin's AI assistant — the AI launcher icon)
    1. **AI App Creation**
        a. From the workspace (any page other than Modules Management / Features Management / Builder Tool), open Cavin's AI and select the **Create** tab.
        b. Verify the helper line reads "Create builds a new application from your description".
        c. Describe the application to build and send. Verify creation progress is shown and "Application created from AI." appears on completion.
        d. Verify the generated application, its modules and features are listed and can be opened in the workspace.
        e. If the AI asks clarifying questions, answer them and verify creation continues (no success message is shown until something is actually created).
    2. **AI Module Creation**
        a. Open an application's **Modules Management** page, open Cavin's AI and select the **Create** tab.
        b. Verify the helper line reads "Create builds a new module here".
        c. Describe the module to build and send. Verify "Created N module(s) from AI." appears and the module list refreshes with the new module(s).
        d. Open the created module and verify it can be edited further.
    3. **AI Feature Creation**
        a. Open a module's **Features Management** page (or the Builder Tool), open Cavin's AI and select the **Create** tab.
        b. Verify the helper line reads "Create builds a new feature here".
        c. Describe the feature to build and send. Verify "Created N feature(s) from AI." appears and the feature list refreshes.
        d. From the Builder Tool, verify you are taken to the first created feature in the builder.
        e. Verify the generated form has appropriate components/fields, then publish it and verify records can be created, viewed and edited.
    4. **Create / Ask mode behaviour**
        a. In the Create tab, type a question instead of a build request. Verify nothing is created and an "Ask this in Ask mode" suggestion is offered; clicking it moves the question to the Ask tab.
        b. With a user who has only AI Builder or only AI Ask access, verify only that tab is available; with no create access, verify past conversations are read-only.
        c. Verify conversation history can be searched and reopened for each mode.

5. **AI Ask FAQ**

   Scope: AI Ask FAQ is trained only on the following platform documentation. Questions must be asked on these topics only:

   - CADP User Login Guide
   - CADP Functional Specifications
   - CADP Erudite Management Complete
   - CADP Workspace Guide
   - CADP User Preferences Guide
   - CADP Role Management
   - AI FAQ Guide Transactions Runtime
   - CADP Login Complete
   - CADP Erudite Management Complete Guide
   - CADP Features Guide
   - CADP Modules Guide
   - CADP Platform Architecture Report
   - CADP Templates Integrations Workflows Basics Guide
   - CADP Features And Submodules Guide
   - CADP Organization Guide
   - CADP Builder Tool Guide

   a. Open Cavin's AI, select the **Ask** tab and ask a question about platform configuration on one of the trained topics above.

   b. Verify a relevant and accurate answer is returned.

   c. Ask a follow-up or ambiguous question and verify it is handled gracefully.

   d. Ask a question outside the trained topics and verify AI Ask FAQ declines or returns an appropriate out-of-scope fallback rather than a fabricated answer.

   e. Verify Ask mode changes nothing in the workspace (it only answers).

- **User Management**
  1. Create/Manage Users
  2. Create/Manage Groups
  3. Import/Export Users
  4. Archive/Unarchive Users
  5. Locked Users
  6. Waiting Approval
  7. Please update existing users in each workspace.
  8. In the Users tab, when creating a new user, the Approval Type default is No Approval.

- **User Preferences**
    1. Profile update – first name, last name updating, profile picture updating.
    2. Change password
    3. Locale and Date-time updating
    4. Interface
        i. Theme change, color change. Color change here will override the brand color for that specific user of that workspace only.
        ii. Records per page is reflecting.
        iii. Language (currently only English accepted)
    5. Customization – module and features are visible based on visibility setting, quick access for that enabled feature is redirecting to feature transactions (records) page.
    6. Privacy & Security
    7. Miscellaneous – Default landing page, preferred currency, auto-save drafts, save interval
    8. Preferred user workspace
    9. Reset Preferences
- **Groups**
    1. Archive/Unarchive

- **Roles**
    1. Platform Roles
    2. Archive/Unarchive
    3. Workspace Role
    4. Feature Access Level
    5. Data Access Level
    6. User or Group Role Mapping
- **Workspace Management**
    1. Create Workspace
    2. Clone Workspace
    3. Archive Workspace
    4. Workspace Settings
- **Organization Management**
    1. Organization Details
    2. Organization Address
    3. Branding Details
- **Module/Feature Management**
    1. Create/Manage Module
    2. Create/Manage Features
    3. Design components for features
    4. Features and Modules can be published
    5. Audit Logs (Platform & Workspace)
    6. Version Logs (Workspace & Workflow)
    7. Staging
    8. Feature Card and Module Card Customization
- **Notification & Alerts**
    1. Email Notification & Templates – Create template and use template to send notification
    2. SMS Notification & Templates – Create template and use template to send notification
    3. WhatsApp Notification & Templates – Create template and use template to send notification
    4. In-App Notification & Templates – Create template and use template to send notification
    5. Push Notification & Templates – Create template and use template to send notification

- **Erudite Management**
  1. **Configuration (prerequisite)**
     a. Enable Erudite Management in Role Management before testing.
     b. Verify users without the role cannot access the Erudite Management module.
  2. **FAQs**
     a. Create a new FAQ with a question, answer, and category (Account, Billing, General, Technical).
     b. Verify the FAQ appears in the FAQ list after saving.
     c. Enable "Display this on the dashboard" and verify the FAQ is visible on the dashboard.
     d. Edit an existing FAQ and verify changes are saved correctly.
     e. Delete a FAQ and verify it is removed from the list.
     f. Search for a FAQ by keyword and verify correct results are returned.
  3. **Documentation**
     a. Upload a document (PDF or DOC, max 5 MB) with a name, description, and category.
     b. Verify the document appears in the Documentation list after saving.
     c. Attempt to upload a file exceeding 5 MB and verify a validation error is shown.
     d. Attempt to upload an unsupported file format and verify it is rejected.
     e. Enable "Display this on the dashboard" and verify the document is visible on the dashboard.
     f. Edit document details and verify changes are saved correctly.
     g. Delete a document and verify it is removed from the list.
  4. **Tutorial Videos**
     a. Upload a video (MP4, MOV, or MKV, max 50 MB) with a name, description, and category.
     b. Verify the video appears in the Tutorial Videos list after saving.
     c. Attempt to upload a file exceeding 50 MB and verify a validation error is shown.
     d. Attempt to upload an unsupported file format and verify it is rejected.
     e. Enable "Display this on the dashboard" and verify the video is visible on the dashboard.
     f. Edit video details and verify changes are saved correctly.
     g. Delete a video and verify it is removed from the list.
  5. **End User Access**
     a. Verify end users can browse and read FAQs, documentation, and tutorial videos.
     b. Verify items marked "Display on dashboard" appear on the user's dashboard.
     c. Verify items not marked for dashboard display do not appear on the dashboard.

- **Components**
    1. Layouts
        1. Column
        2. Section (various nested scenarios)
    2. Data Management (API Integration, Device Integration)
    3. Input Components (Phone Number, Password, Dropdown with Feature as Field Value)
    4. Specialized Input & Display
    5. Display and Interaction
    6. Media and Files
    7. Forms and Validation
    8. Utility Components
    9. Data Config Component
    10. Hyper Link Component
    11. Favorite Component
    12. OCR Component
        a. Add the OCR component to a feature form and upload an image or document.
        b. General text extraction → verify the text is extracted from the uploaded image/document and displayed within the selected output textarea
        c. Verify the extracted text is saved correctly with the record and is visible on View/Edit.
    13. Analytics & Report Components
        1. Chart Component
        2. Gantt Chart Component
        3. KPI Card Component
    14. Form UI
        1. Table component and Table level formulas, Summary row formulas
    15. Reusable Component
- **Page Builder**
    1. Create a new page and verify it is saved and listed.
    2. Drag-and-drop layouts (columns, sections) and reusable components onto the page canvas.
    3. Configure component properties and verify changes reflect on the page.
    4. Reorder, move, and delete components on the canvas.
    5. Edit an existing page and verify changes are persisted.
    6. Delete a page and verify it is removed from the list.
    7. Publish the page and verify it renders correctly at runtime.
    8. Verify the page is responsive across mobile, tablet, and desktop.
    9. Verify role-based access — only authorized users can build/edit pages.

- **Workflow Management**
    1. Visual canvas for creating workflows
    2. Drag-and-drop node placement
    3. Node connection with visual links
    4. Canvas zoom and pan controls
    5. Grid snap for alignment
    6. Create new workflows
    7. Edit existing workflows
    8. Delete workflows
    9. List/view all workflows
- **Workflow Nodes – Control Flow**
    1. Start Node (workflow trigger point)
    2. End Node (workflow termination)
    3. Condition/Decision Node (if-else logic)
- **Workflow Nodes – Communication**
    1. Email Notification Node
    2. SMS Notification Node
    3. User Task/Approval Node
    4. System Notification Node
- **Workflow Monitoring**
    1. Execution history logs
    2. Workflow instance status tracking
    3. Node-level execution details
    4. Error logs and stack traces
    5. Approval node changes for user selection
    6. Added SLA and escalation reminder for approvals
    7. Added parallel workflow approval node
- **Printing Templates**
    1. QR Code – Generate and Print
    2. Barcode – Generate and Print
    3. Invoice – Generate and Print
- **Role-Based Dashboards**
    1. Role-based dashboard template creation
    2. Dashboard rendering based on created template

- **Scheduler & Task Scheduler**
    1. Create and use Schedule
    2. Create and use Task Schedule
    3. Event-based task schedule
    4. Schedule-based task schedule
    5. Schedule a notification
    6. SMS
    7. Mail
    8. WhatsApp
    9. Push
    10. In-App
    11. Schedule a report send (PDF) through mail from a configured email template.
- **Formula Builder Functionality**
    1. Same Feature
        a. Arithmetic
        b. Logical
        c. Comparison
        d. Text
        e. Math
        f. Date
        g. Conditional
        h. Miscellaneous
        i. Aggregate → with filter conditions and without filter conditions.
    2. Cross Feature
        a. Arithmetic
        b. Logical
        c. Comparison
        d. Text
        e. Math
        f. Date
        g. Conditional
        h. Miscellaneous
        i. Aggregate → with filter conditions and without filter conditions.
    3. Incompatible datatype operations, for example, arithmetic on a text input field, should show a validation error.
    4. Unrecognized fields, functions, or operators should show a validation error.
    5. If external fields are called without where() conditions, it should show MULTIPLE RECORDS FOUND ERROR in preview for non-aggregate functions/operations.
    6. Variables are not working. Review which variables should be displayed in the modal.

- **Formula Builder UI**
    1. On clicking a feature from the feature list, fields in the selected feature should be displayed, except for the current field, to prevent self-call loops.
    2. On selecting external feature fields, where() should appear in the modal.
    3. On selecting where(), the modal should automatically switch to the Feature and Fields tab.
    4. On hovering over a field, field information such as module name and feature name should be displayed.
    5. The Clear button or backspace should work as follows:
        i. First click of clear/backspace will select the token.
        ii. Next click of clear/backspace will clear that token.
    6. Empty formulas should not be saved.
    7. Formulas should not be saved without validation.

- **Formula Builder – Helper Tools**
  1. **Cheat Sheet** (**?** icon)
     a. Open the Cheat Sheet and verify functions are grouped by category (Lookup, Nested Lookup, Aggregation, Conditional, Math, String, Date).
     b. Search for a function and verify the list filters correctly.
     c. Click the copy icon on an example and verify it is copied to the clipboard.
     d. Press Esc and verify the modal closes.
  2. **Examples**
     a. Open the Examples drawer and verify 50 scenarios across all categories are listed.
     b. Filter by a category and search; verify results update.
     c. Click "Use This Pattern" and verify the formula is inserted into the editor.
  3. **Test Formula (Playground)**
     a. For a current-feature calculation (e.g. *{number_field} * 5*), open Test Formula, verify a type-aware input appears per field (number/date/true-false/text), enter values, run, and verify the correct result, type, and (where relevant) matched-record count.
     b. For a lookup formula (uses *.where()* / external feature fields), verify NO value-input form is shown and a "Lookup formula – test against a real record" message appears instead.
     c. For an aggregate formula (SUM/COUNT/AVG/MIN/MAX of records), verify the "Aggregate formula – test against a real record" message is shown and the form is not available.
     d. Verify the Test Formula button is disabled when the formula is empty.
  4. **Template Library**
     a. Open the Template Library and verify the **Most Used** tab lists the 20 platform built-in templates (e.g. GST, TDS, net salary, lookups, aggregations).
     b. Search and filter by category; verify results update.
     c. Click "Use This Template" and verify the formula is inserted into the editor.
     d. With a formula in the editor, click "Save current formula as template", fill title/category/description, save, and verify it appears under **My Templates**.
     e. Delete a saved (non built-in) template and verify it is removed. Verify built-in templates cannot be deleted.
  5. **Formula Wizard**
     a. **Lookup mode:** choose "Lookup a value", select a source feature, select the field to return, add a match condition (verify the "Right side is a field reference" option shows a dropdown of the current feature's fields), check the live preview, and Insert.
     b. **Aggregate mode:** choose "Aggregate records", select feature/field, optionally add one filter condition, choose the function (SUM/COUNT/AVG/MIN/MAX), and Insert.
     c. Verify the formula preview only appears after a return field is chosen, and that going Back keeps your previous selections.
     d. After inserting, hover over the tokens in the editor and verify the correct Module name and Feature name are shown for each (including the source feature for the looked-up field).

6. **Smart Validation Messages**
    a. Enter an invalid formula (e.g. unknown function, missing where(), wrong argument count) and click Validate Expression; verify the error includes a helpful suggestion and an example, not just a generic message.

- **Integrations**
    1. Step 1: Go to the Integration Type field and select "Third-Party Software".

       Expected Result: A new dropdown field called Select Type should appear.
    2. Step 2: Check the options inside the Select Type dropdown.

       Expected Result: It should contain options including "Third-Party API" and "Google Drive".
    3. Step 3: Choose "Third-Party API" from the dropdown.

       Expected Result: The configuration fields specific to the Third-Party API should be displayed.
    4. Step 4: Change the selection to "Google Drive".

       Expected Result: The system should dynamically display the Google Drive configuration fields: Client ID, Client Secret, and Refresh Token.

This build — additional test focus:

- Date filters in KPI counts now check each record's own date; ADD_DAYS works in filters — A record aggregate's filter (`COUNT_RECORDS({Task ID}, {Due Date} < TODAY())`, and the; same for SUM / AVG / MIN / MAX and the DISTINCT_* family) now tests every record against its OWN; field values. Filters can also use `ADD_DAYS(<date>, <whole days>)` — `<date>` being `TODAY()`,; `NOW()`, a quoted date or a date field — so ageing buckets such as "overdue 8–15 days" can be; written as `{Due Date} <= ADD_DAYS(TODAY(), -8) and {Due Date} >= ADD_DAYS(TODAY(), -15)`.
- A formula using a temp variable returned 0 — A formula field whose expression was a temp-variable reference - `@field.Amt_formula`; returned `0` with no error, so the field showed `0.00` however much data was on the form. The; identical arithmetic written inline returned the right number.
- Aggregates over the current feature's saved records now honour their filter — A formula that aggregates the CURRENT feature's stored records with a condition —; `COUNT_RECORDS({Product}, {Price} > 30)`, or the `.where()` spelling of the same question —; ignored the condition. It returned `0`, or counted every record, depending on the shape. The; unfiltered forms were always correct, which is what made this look intermittent.
- A "Saved records" field picked by display name failed validation — Picking a field from the current feature's "Saved records" list and using it in an; aggregate was refused by Validate Expression with; `External field "{price_1}" is not mapped to any known feature`, so the formula could not be; saved — even though the engine computed it correctly at runtime.
- Designer screens show field names for charts saved by field ID — While designing, the dashboard designer canvas and the report designer canvas now show; a chart field's name. Previously they could show its field ID (for example `557cb79d-…`) for a; chart picked after 2026-11-06. This appeared as the series name or legend of a single-series chart; (single-series charts have no "Series 1 Name" box, so they always use the fallback), as the; donut subtitle and the gauge name, and in the report designer's "X Fields / Y Fields / Category /; Value" pills. Saved dashboards and reports were already showing names.

- Tables choose their fields by ID; report charts show labels for ID fields — A dashboard or report table now picks its columns by field ID (the builder already; saves `selectFieldIds` next to `selectFields`), falling back to the saved name only when a column; was saved without a field ID — so tables built before this keep every column. Report charts; saved with field IDs show field labels in legends, measure names, the X-field tag and the donut; subtitle (the report service translates the IDs for grouping — see the client-ws change log).
- Dashboard charts and tables work on workflow-created records — Dashboard charts now save the fields you pick by field ID (the dropdowns still show the; field names), and dashboard charts and tables read records by that ID. Charts on features whose; records are created by a workflow — Task Management, RCF History Report — now show their data; instead of "No data available" or one "Other" slice, and tables show all their columns.
- A formula that is only a temp-variable reference never recalculated — Selecting an expression, saving it as a temp variable, then replacing the expression; with that variable - `@field.Amt_formula` - left the field showing `0.00` forever. The same; expression written inline in a field beside it updated normally, which is what made this read; as "the temp variable does not work".

10. What to Test on Mobile

- Login – Normal Login
- Login – SSO (Sign In with Google, Sign In with Microsoft)
- Workspace Switch
- Module List
- Feature List
- Components
- Feature Data – Create, View, Edit, Clone, Delete
- Dashboard
- Integration – Weighing Machine, Third-Party API (Budgie, SAP), and Payment Gateways (Stripe, Razorpay, and PayU)
- Theme
- Push Notification
- Workflow – Approve and Reject
- Charts – Feature Data
- Data Table – Feature Data
- Security – Prevent copy, paste, cut, and text selection